

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch: Chemical Semester: V Subject Code & Name: BTCHC501-Mass Transfer Operations - I Max Marks: 60 Date: 28/01/23 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	State Fick's law of mass diffusion. Explain equilibrium phenomena in gas - liquid mass transfer operation.	Understanding	6
B)	Define mass transfer coefficient? What are the different mass transfer theories to predict the mass transfer coefficient?	Applying	6
C)	In an oxygen–nitrogen gas mixture at 1 atm. 25°C, the concentrations of oxygen at two planes 0.2 cm apart are 10% and 20% (by mole) respectively. Calculate the flux of oxygen in kmol/m ² s when (i) nitrogen is non-diffusing (ii) there is equimolar counter diffusion. Diffusivity of oxygen in nitrogen is 0.215 cm ² /s. Gas Constant R= 8314 Pa m ³ /(kmol K)	Evaluating	6
Q.2	Solve Any Two of the following.		12
A)	Write down material balance for single stage operation and two stage counter-current and crosscurrent operation for general mass transfer operation.	Applying	6
B)	Write down procedure (with graphical depiction) for counter-current cascade operation of gas absorption tower when (i) individual stage is ideal and co-current (ii) individual stage is ideal and countercurrent.	Applying	6
C)	Define ideal stage. Write down the design procedure to determine the height of the transfer units (HTU) needed for given absorption or adsorption operation.	Evaluating	6
Q. 3	Solve Any Two of the following.		12
A)	Discuss the selection criteria of solvent in detail for gas absorption process.	Understanding	6
B)	List out some important industrial gas-liquid absorption processes. What are the important parameters to design gas absorption tower? What is mean by Absorption Factor?	Applying	6
C)	An air–NH ₃ mixture containing 5% NH ₃ is being scrubbed counter currently with water in a packed tower to recover 95% NH ₃ . Total gas inlet flow rate was 3000 kg/ m ² h, and pure water inlet flowrate was 2500 kg/ m ² h. The equilibrium relation is given by y* = 0.98x, where x and y are mole fraction units. Determine the ammonia's concentration in the exit water stream.	Evaluating	6
PTO			

Q.4	Solve Any Two of the following.		12
A)	Explain in detail the selection criteria of solvent for extraction system.	Understanding	6
B)	Explain following extraction/leaching equipment: a) Mixer Settler unit, b) Any moving bed leaching equipment	Understanding	6
C)	Explain the procedure to find number of ideal stages for continuous countercurrent Type I extraction system when (i) actual solvent rate is given (ii) actual solvent rate is based on minimum solvent rate.	Understanding	6
Q. 5	Solve Any Two of the following.		12
A)	Explain Langmuir isotherm for adsorption in brief. Explain applications of Freundlich equation for single stage and two stage adsorption process with example.	Understanding	6
B)	Explain the design and working principle of any two gas dispersed units out of bubble column, mechanical agitated bubble column or tray towers.	Understanding	6
C)	The equilibrium decolourisation data for a certain system using activated carbon is given by the equation, $Y = 0.004 X^2$ where Y is g colouring impurity/kg impurity free solution and X is g colouring impurity/kg pure activated carbon. Calculate the amount of activated carbon required per 1000 kg of impurity free solution to reduce the impurity concentration from 1.2 to 0.2 g/kg of impurity free solution using a single stage operation.	Evaluating	6
*** End ***			